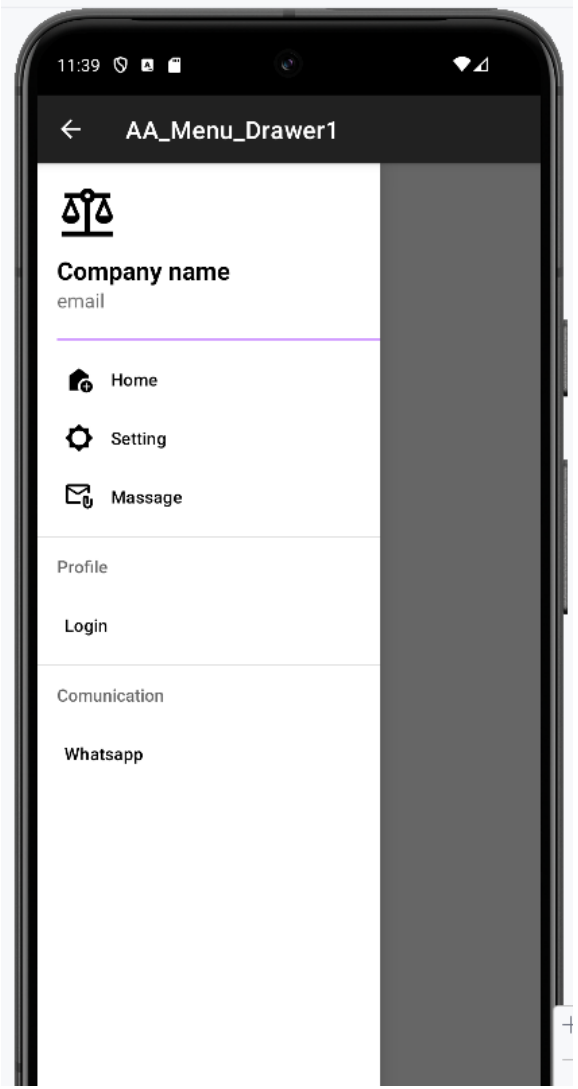
13. Drawer Menu

Android Studio

Android Navigation Drawer Menu Material Design

1. Add library to build.gradle
2. Add header resource file
3. Add menu folder
4. Add Icons
5. Add groups



Adding a dependency to the project

```
implementation 'com.google.android.material:material:1.3.0-alpha03‘
```

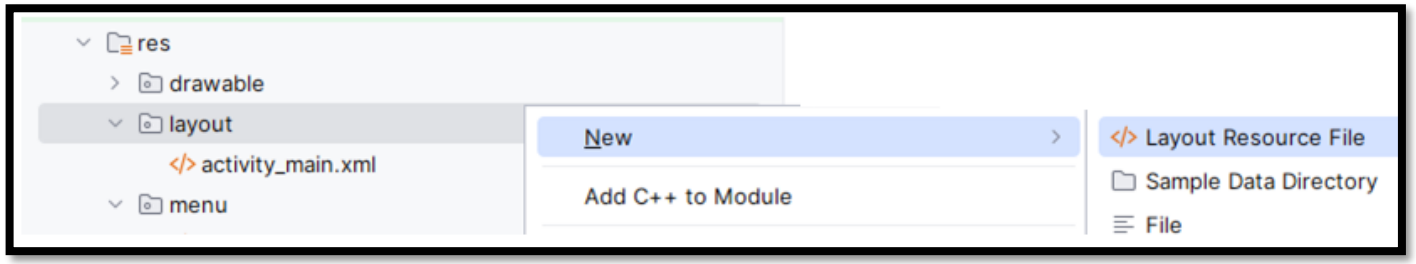
3 Sync Now Ignore these changes

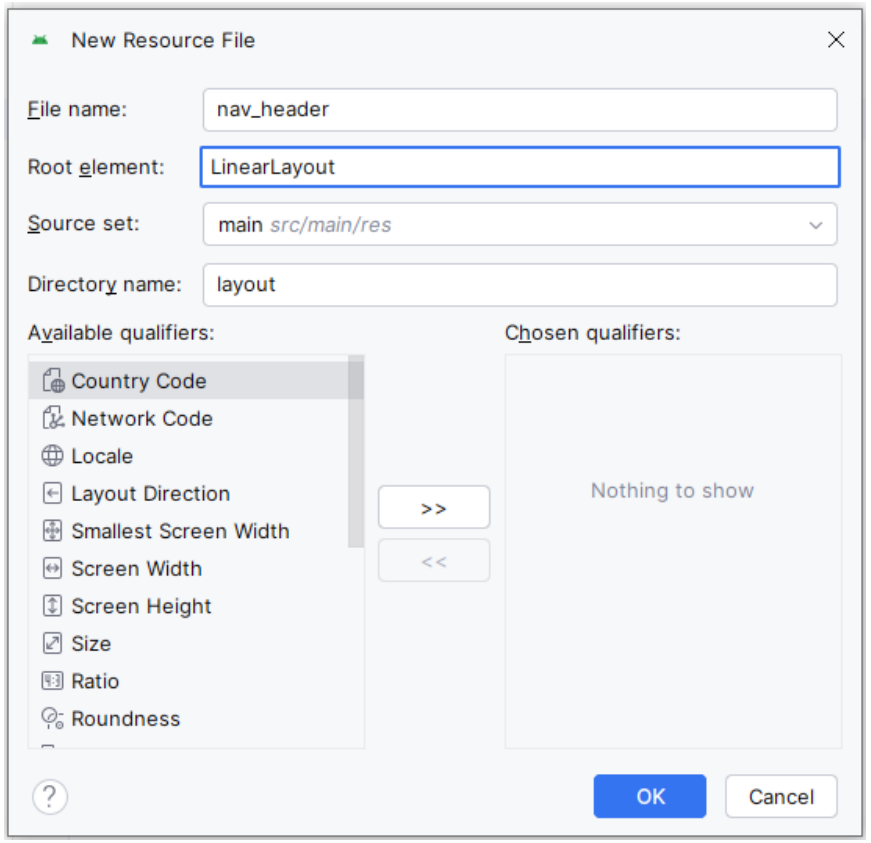
Gradle Scripts
- build.gradle.kts (Project: AA_Menu_Drawer1)
- build.gradle.kts (Module :app) 1
- proguard-rules.pro (ProGuard Rules for ":app")
- gradle.properties (Project Properties)
- gradle-wrapper.properties (Gradle Version)
- libs.versions.toml (Version Catalog "libs")
- local.properties (SDK Location)
- settings.gradle.kts (Project Settings)

```
38  }
39
40  dependencies {
41      implementation(libs.androidx.core.ktx)
42      implementation(libs.androidx.appcompat)
43      implementation(libs.material)
44      implementation(libs.androidx.activity)
45      implementation(libs.androidx.constraintlayout)
46      testImplementation(libs.junit)
47      androidTestImplementation(libs.androidx.junit)
48      androidTestImplementation(libs.androidx.espresso.core)
49  2   implementation("com.google.android.material:material:1.3.0-alpha03‘")
50  }
```

Add Header in Layout folder

- Write click on "**layout**"
- New → **Layout Resource File**
- File Name → **nav_header**
- Root element → LinearLayout
- **Press → OK**
- **Open nav_header then add the code next slie**





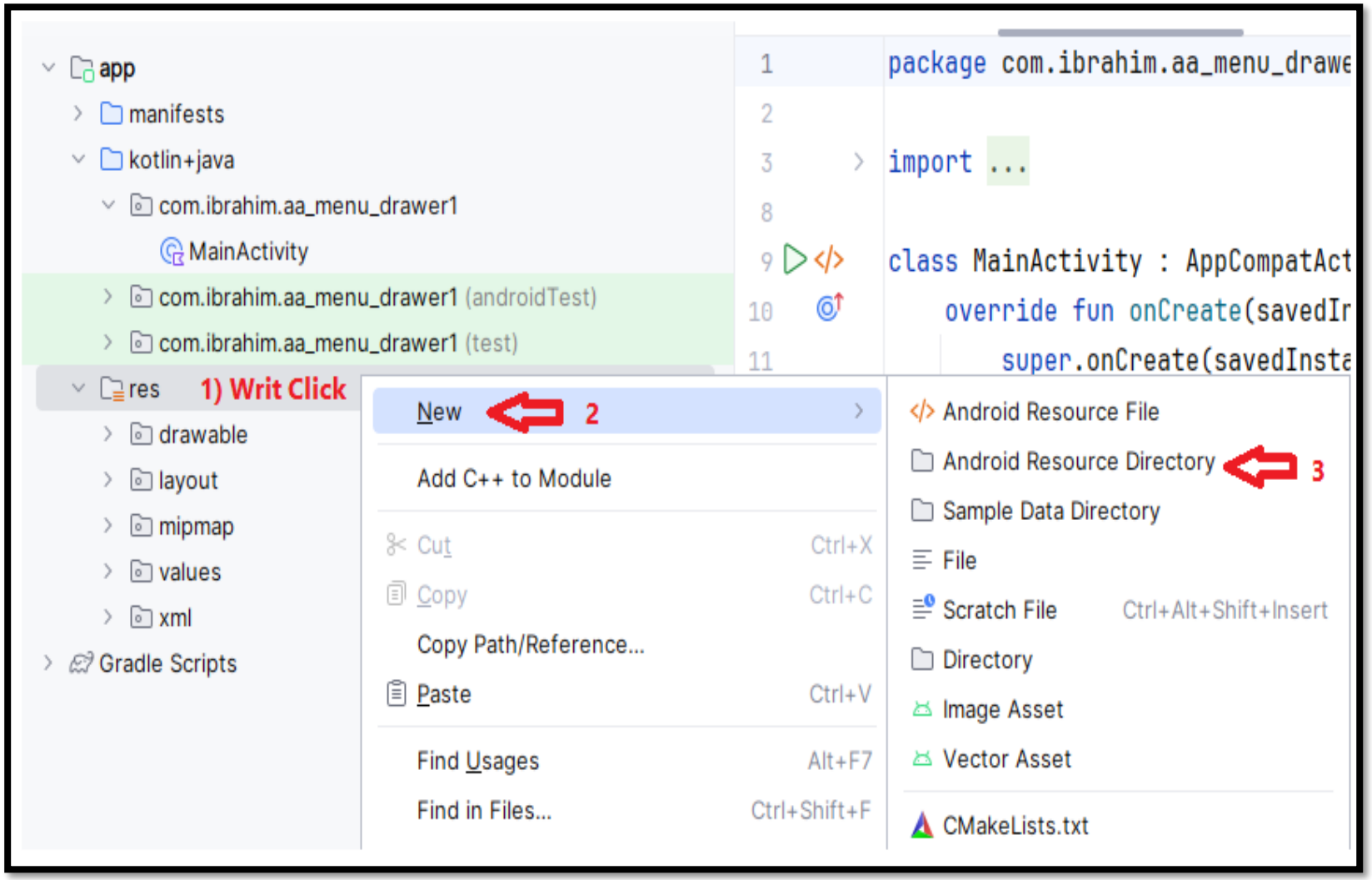
Add design to nav_header.xml

nav_header.xml

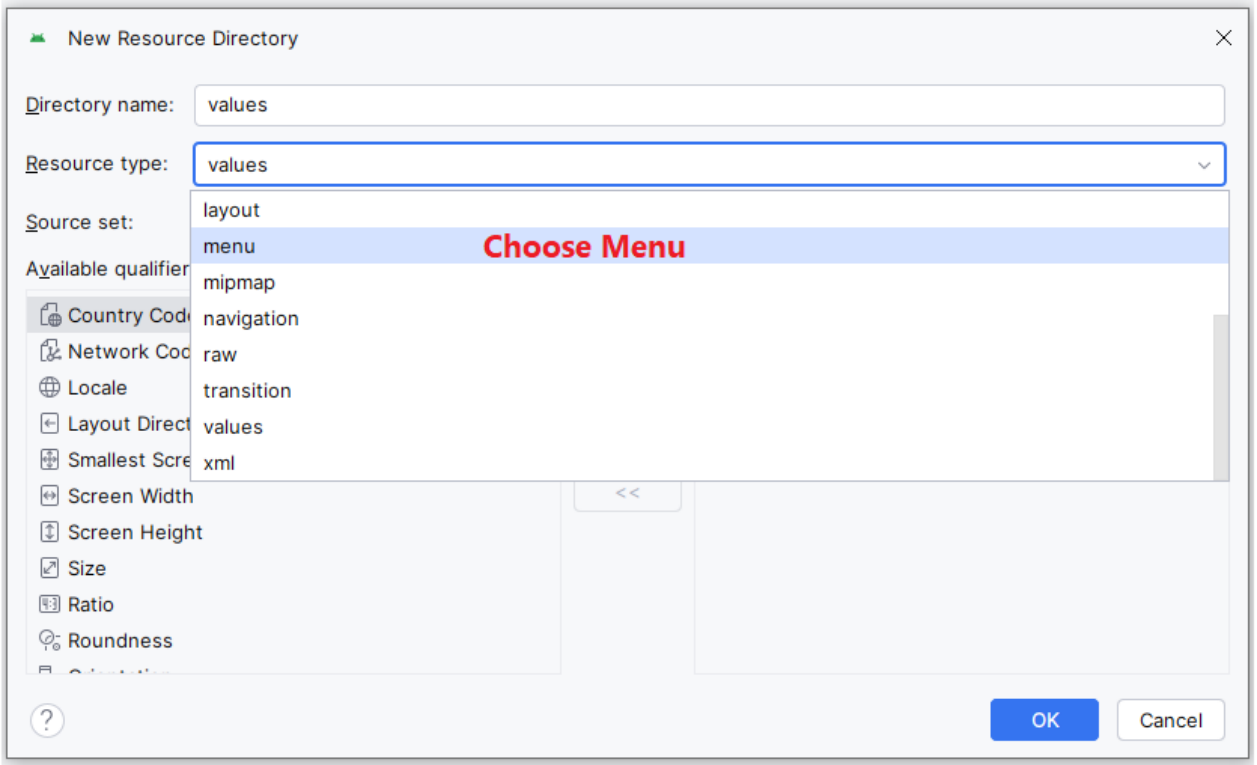
```xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="50dp"
        android:layout_height="50dp"
        tools:srcCompat="@tools:sample/avatars" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:textColor="@color/black"
        android:paddingTop="8dp"
        android:textStyle="bold"
        android:text="Company name" />
    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="16dp"
        android:text="email" />
    <View
        android:layout_width="match_parent"
        android:layout_height="2dp"
        android:layout_marginTop="20dp"
        android:background="#D099FF"/>
</LinearLayout>
```

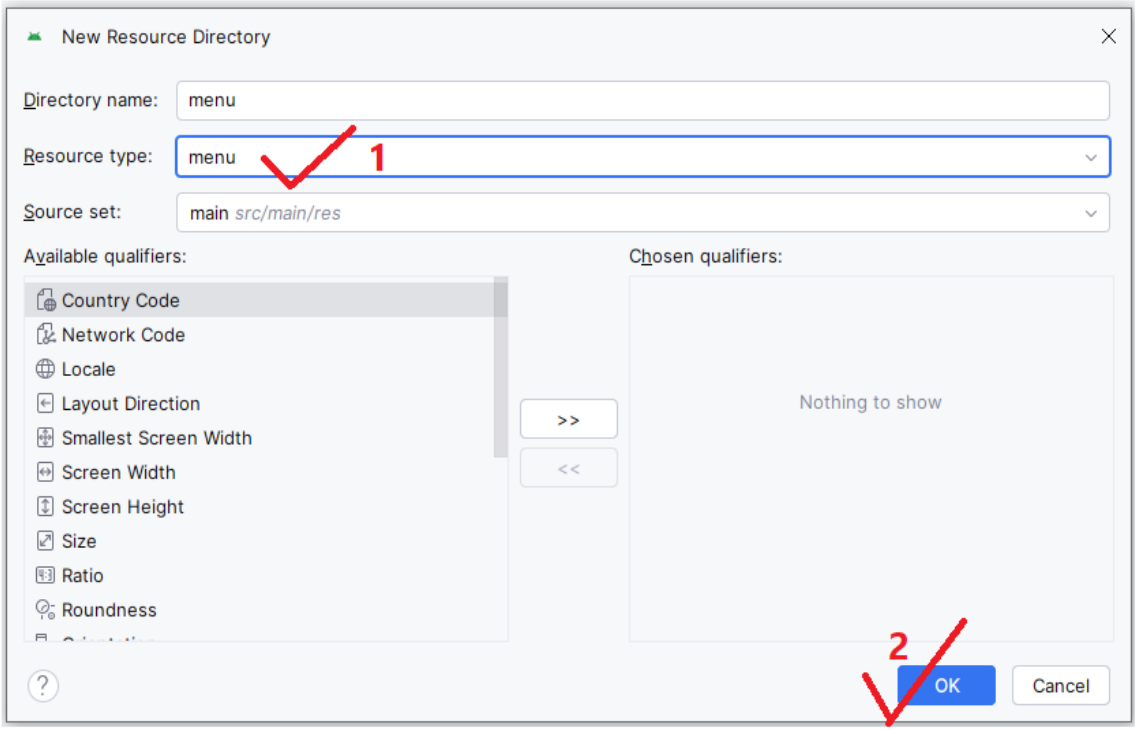
Add menu folder to resource

- Write click on "**res**"
- New → **Android Resource Directory**
- Menu → **Ok**



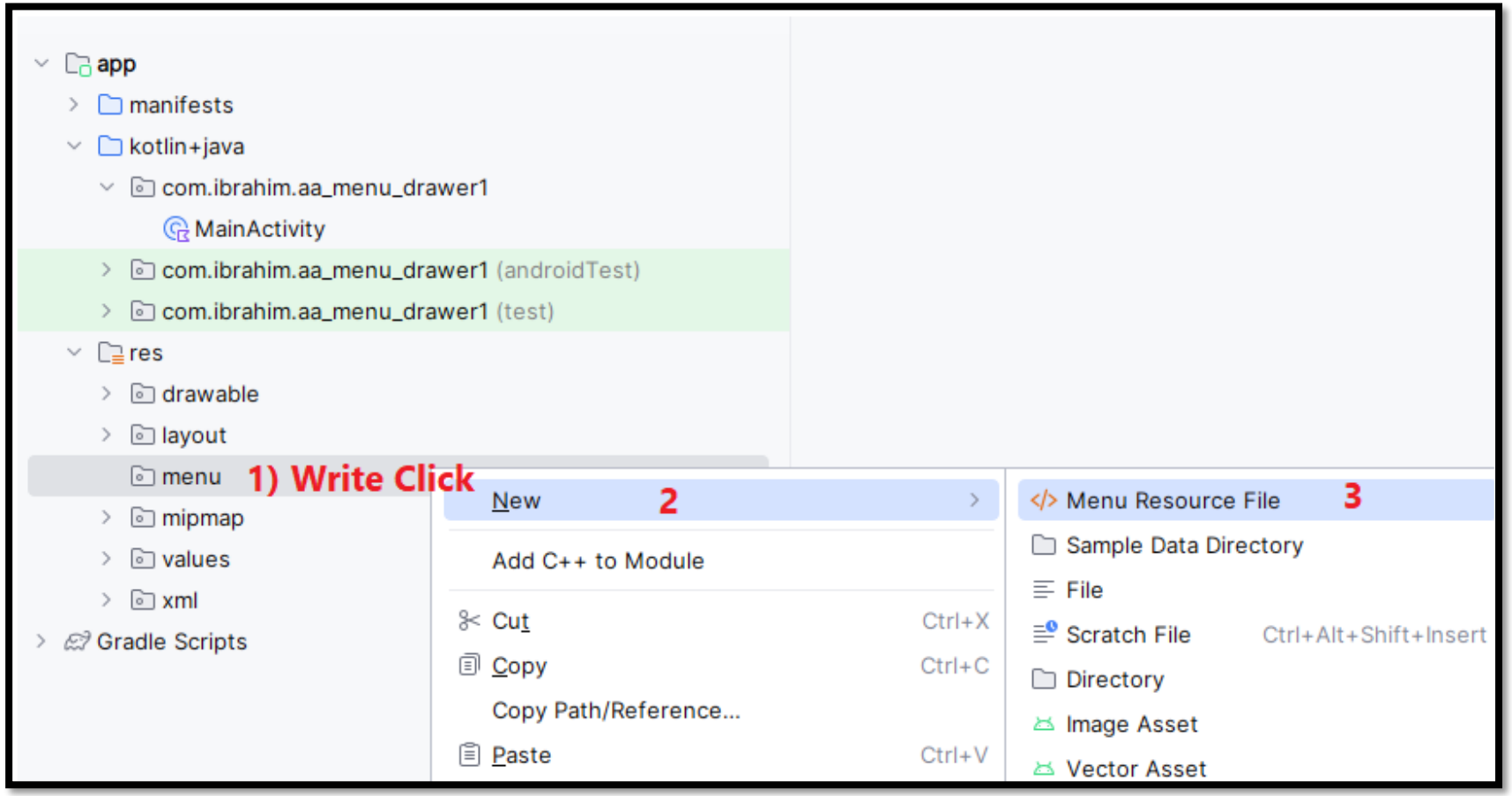
Choose menu


New Resource Directory
Directory name:
values
Resource type:
values
Source set:
Available qualifier
layout
menu
Choose Menu
mipmap
navigation
raw
transition
values
xml
Country Cod
Network Coc
Locale
Layout Direct
Smallest Scre
Screen Width
Screen Height
Size
Ratio
Roundness
<<
OK
Cancel

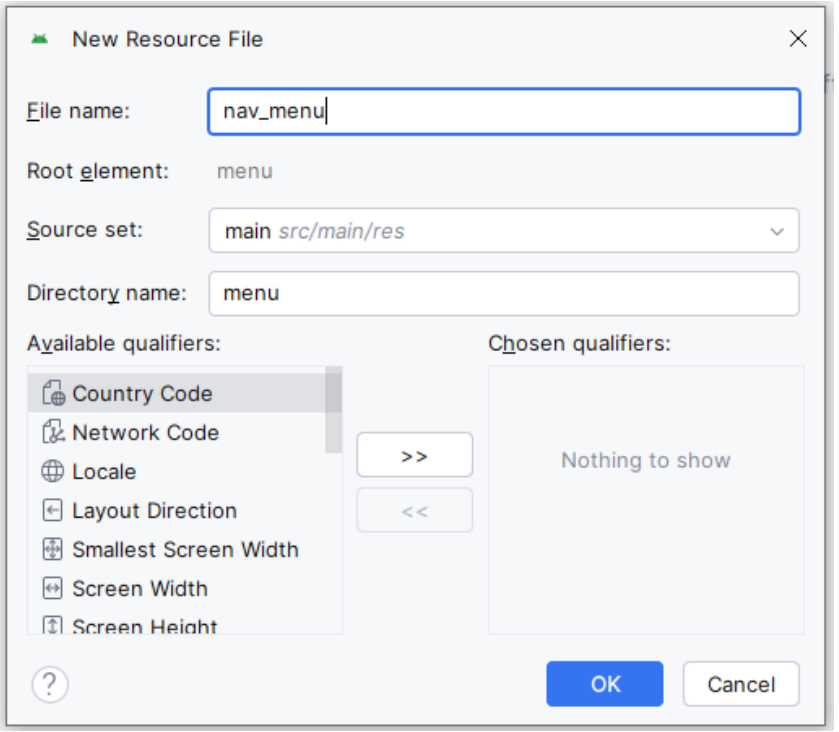

New Resource Directory
Directory name:
menu
Resource type:
menu
1
Source set:
main src/main/res
Available qualifiers:
Chosen qualifiers:
Country Code
Network Code
Locale
Layout Direction
Smallest Screen Width
Screen Width
Screen Height
Size
Ratio
Roundness
>>
<<
Nothing to show
2
OK
Cancel

Add file to menu folder

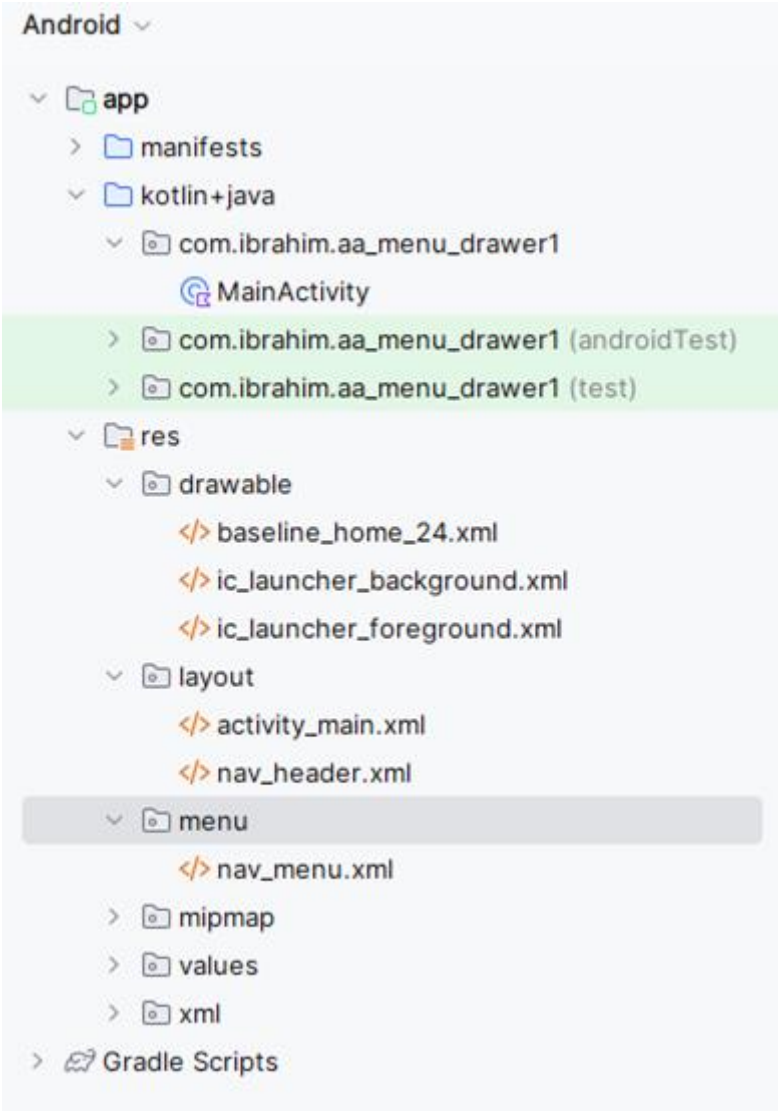
- Write click on “menu”
- New → **Menu Resource File**
- File Name → **menu** → **Ok**
- Add nav_menu



nav_menu.xml:


New Resource File
File name:
nav_menu
Root element:
menu
Source set:
main src/main/res
Directory name:
menu
Available qualifiers:
Chosen qualifiers:
Country Code
Network Code
Locale
Layout Direction
Smallest Screen Width
Screen Width
Screen Height
>>
<<
Nothing to show
OK
Cancel

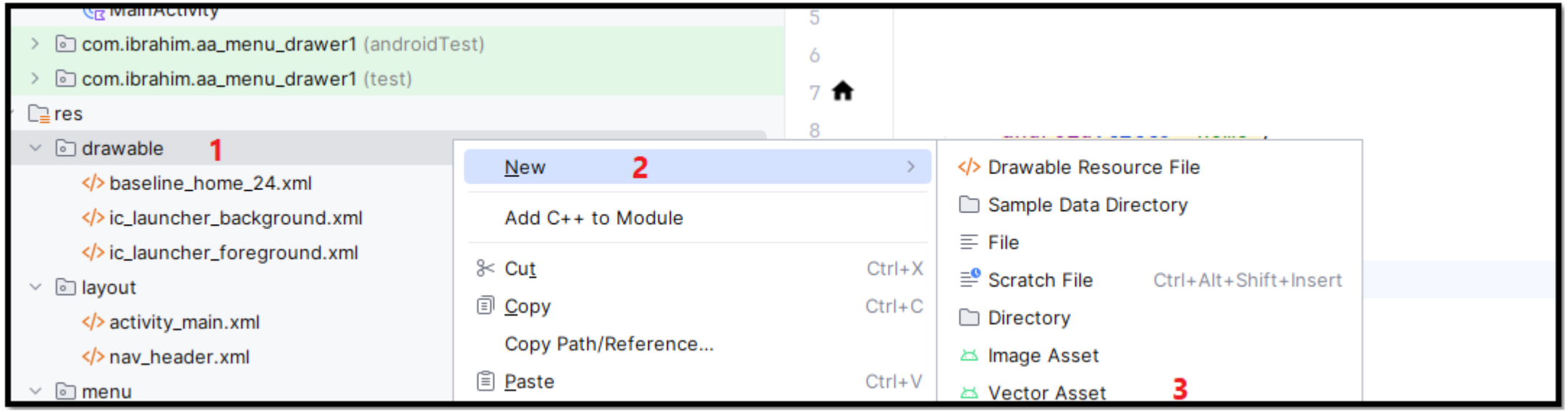
nav_menu.xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
</menu>

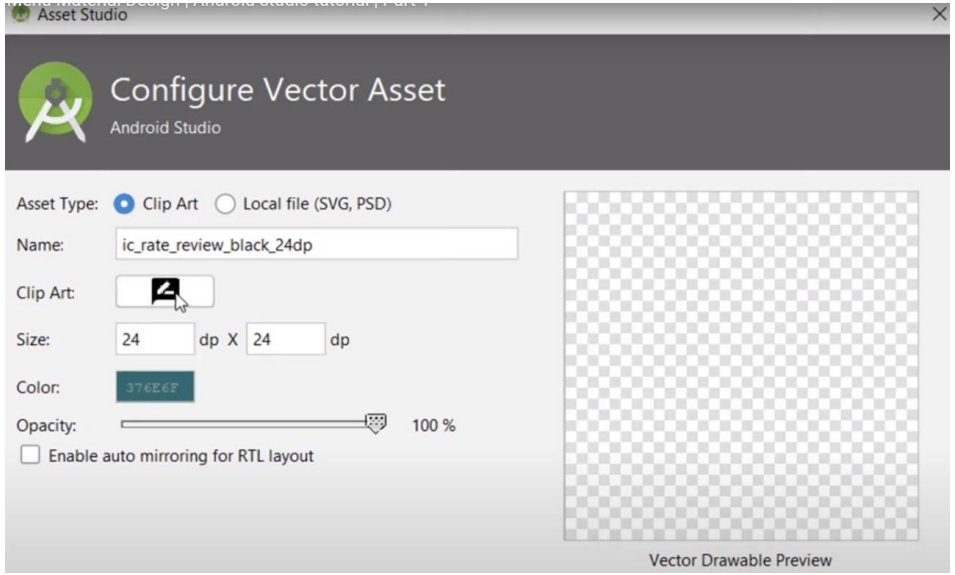

Android
app
manifests
kotlin+java
com.ibrahim.aa_menu_drawer1
MainActivity
com.ibrahim.aa_menu_drawer1 (androidTest)
com.ibrahim.aa_menu_drawer1 (test)
res
drawable
baseline_home_24.xml
ic_launcher_background.xml
ic_launcher_foreground.xml
layout
activity_main.xml
nav_header.xml
menu
nav_menu.xml
mipmap
values
xml
Gradle Scripts

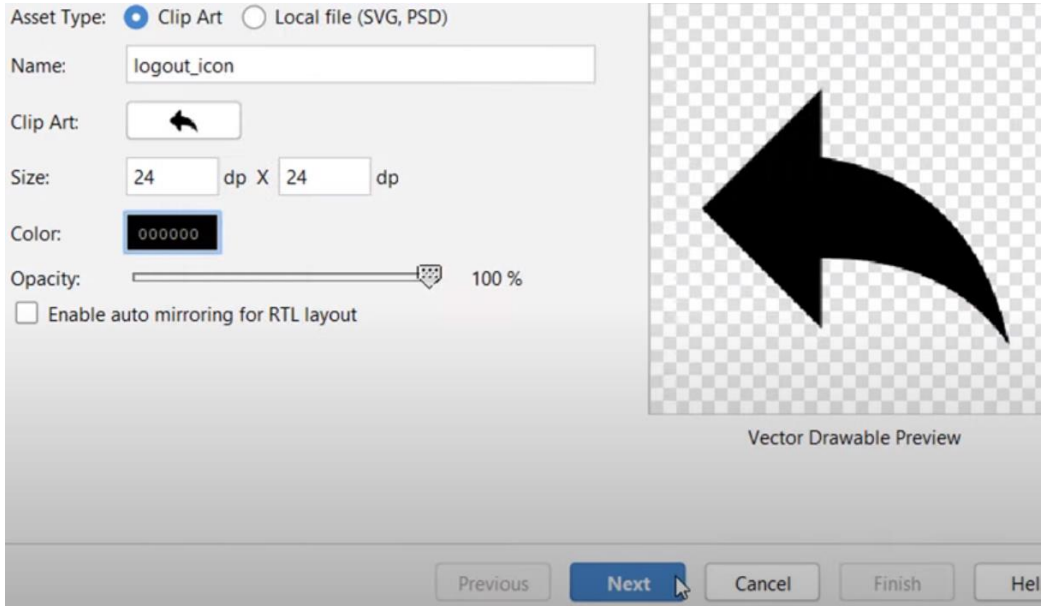
Add new Icon to drawable folder

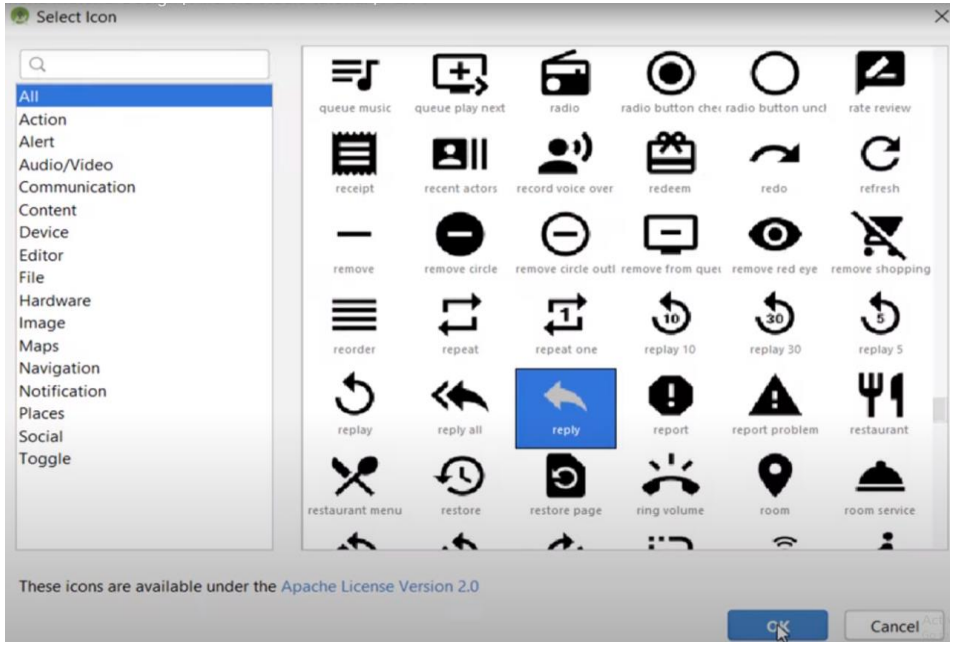
- Write click on “**drawable**”
- New → **Victor Asset**
- Clip Art → **Select Icon**
- Change: Name = **home_icon**
- Color = **Black**
- Press **Finish**
- **Add All Icons**

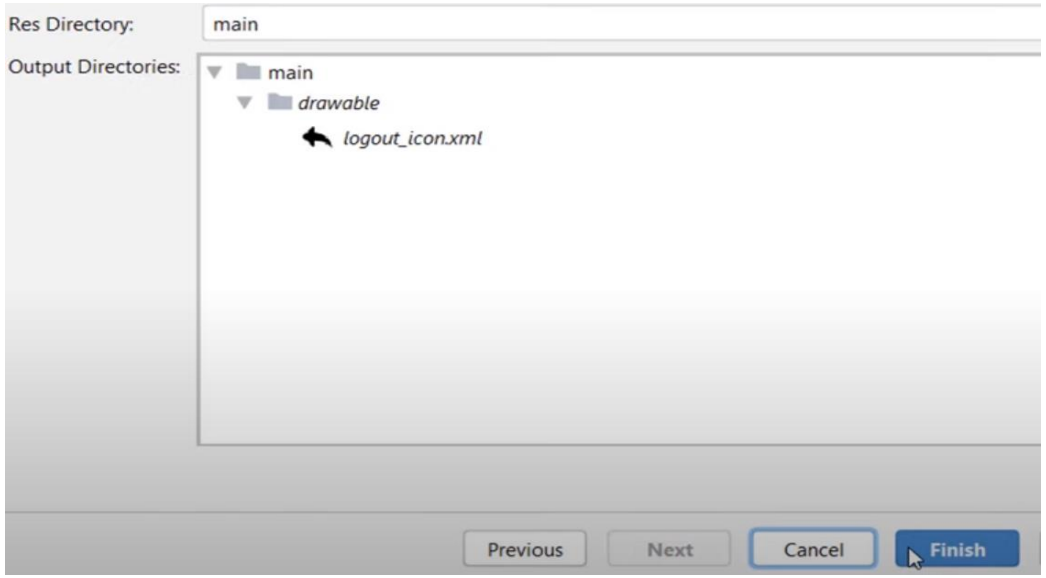
Before build the navigation menu file you should select all icons for menu:




Asset Studio
Configure Vector Asset
Android Studio
Asset Type:
Clip Art
Local file (SVG, PSD)
Name:
ic_rate_review_black_24dp
Clip Art:
Size:
24
dp X
24
dp
Color:
376E6F
Opacity:
100 %
Enable auto mirroring for RTL layout
Vector Drawable Preview


Asset Type:
Clip Art
Local file (SVG, PSD)
Name:
logout_icon
Clip Art:
Size:
24
dp X
24
dp
Color:
000000
Opacity:
100 %
Enable auto mirroring for RTL layout
Vector Drawable Preview
Previous
Next
Cancel
Finish
Hel


Select Icon
All
Action
Alert
Audio/Video
Communication
Content
Device
Editor
File
Hardware
Image
Maps
Navigation
Notification
Places
Social
Toggle
queue music
queue play next
radio
radio button che
radio button uncl
rate review
receipt
recent actors
record voice over
redeem
redo
refresh
remove
remove circle
remove circle outl
remove from que
remove red eye
remove shopping
reorder
repeat
repeat one
replay 10
replay 30
replay 5
replay
reply all
reply
report
report problem
restaurant
restaurant menu
restore
restore page
ring volume
room
room service
These icons are available under the Apache License Version 2.0
OK
Cancel


Res Directory:
main
Output Directories:
main
drawable
logout_icon.xml
Previous
Next
Cancel
Finish

Android

- app
  - manifests
  - kotlin+java
    - com.ibrahim.aa_menu_drawer1
      - MainActivity
    - com.ibrahim.aa_menu_drawer1 (androidTest)
    - com.ibrahim.aa_menu_drawer1 (test)
  - res
    - drawable
      - home_icon.xml
      - ic_launcher_background.xml
      - ic_launcher_foreground.xml
    - layout
      - activity_main.xml
      - nav_header.xml
    - menu
      - nav_menu.xml
    - mipmap
    - values
    - xml
- Gradle Scripts

nav_menu.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    tools:showIn="nav_view">

    <item android:id="@+id/nav_home"
        android:icon="@drawable/home_icon"
        android:title="home" />

</menu>
```

nav_menu.xml

nav_menu.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    tools:showIn="nav_view">

    <item android:id="@+id/nav_home"
        android:icon="@drawable/home_icon"
        android:title="Home" />
    <item android:id="@+id/nav_settings"
        android:icon="@drawable/setting_icon"
        android:title="Setting" />
    <item android:id="@+id/nav_massage"
        android:icon="@drawable/massage_icon"
        android:title="Massage" />

<item  android:title="Profile">
    <menu>
        <item android:id="@+id/nav_login"
            android:title="Login" />
    </menu>
</item>
<item android:title="Comunication" >
    <menu>
        <item android:id="@+id/nav_whatsapp"
            android:title="Whatsapp"/>
    </menu>
</item>
</menu>
```

Design activity_main

nav_header.xml activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/drawerLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <androidx.constraintlayout.widget.ConstraintLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Activity 1"
            android:textSize="30dp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent" />
    </androidx.constraintlayout.widget.ConstraintLayout>

    <com.google.android.material.navigation.NavigationView
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        app:headerLayout="@layout/nav_header"
        app:menu="@menu/nav_menu"
        android:id="@+id/nav_veiw"
        android:fitsSystemWindows="true"
        android:layout_gravity="start"/>

</androidx.drawerlayout.widget.DrawerLayout>
```

Add open & close to strings file

MainActivity.kt strings.xml

Edit translations for all locales in the translations editor.

```
1  <resources>
2      <string name="app_name">AA_Menu_Drawer1</string>
3      <string name="open">Open</string>
4      <string name="close">Close</string>
5  </resources>
```

MainActivity.kt

```kotlin
import android.os.Bundle
import android.view.MenuItem
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.ActionBarDrawerToggle
import androidx.appcompat.app.AppCompatActivity
import androidx.drawerlayout.widget.DrawerLayout
import com.google.android.material.navigation.NavigationView

2 Usages
class MainActivity : AppCompatActivity() {
    4 Usages
    lateinit var toggle: ActionBarDrawerToggle
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
       val drawerLayout : DrawerLayout = findViewById( id = R.id.drawerLayout)
       val navView : NavigationView = findViewById<NavigationView>( id = R.id.nav_view)

        toggle = ActionBarDrawerToggle( activity = this,drawerLayout, openDrawerContentDescRes = R.string.open, closeDrawerContentDescRes = R.string.close)
        drawerLayout.addDrawerListener( listener = toggle)
        toggle.syncState()
```

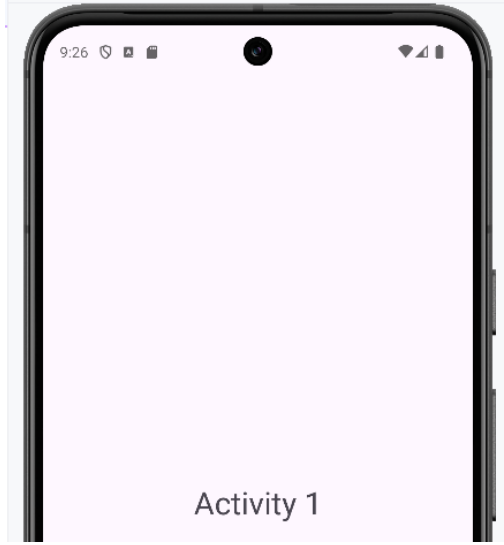
Continues …

```
25          supportActionBar?.setDisplayHomeAsUpEnabled(true)
26          navView.setNavigationItemSelectedListener {
27              when(it.itemId){
28                  R.id.nav_home -> Toast.makeText( context = this,  text = "my home",  duration = Toast.LENGTH_SHORT).show()
29                  R.id.nav_settings -> Toast.makeText( context = this,  text = "my setting",  duration = Toast.LENGTH_SHORT).show()
30                  R.id.nav_login -> Toast.makeText( context = this,  text = "my login",  duration = Toast.LENGTH_SHORT).show()
31
32              }
33              true
34          }
35      }
36      override fun onOptionsItemSelected(item: MenuItem): Boolean {
37          if (toggle.onOptionsItemSelected(item)){
38              return true
39          }
40          return super.onOptionsItemSelected(item)
41      }
42  }
```

Themes file

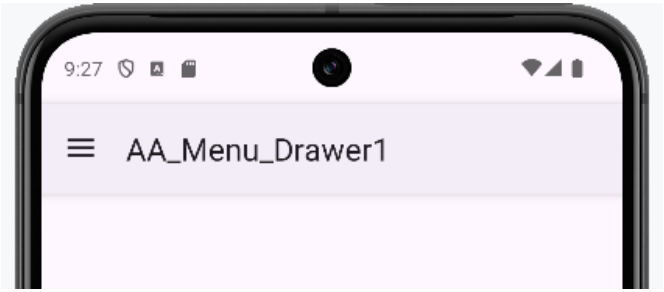
MainActivity.kt themes.xml

```xml
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.AA_Menu_Drawer1" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.AA_Menu_Drawer1" parent="Base.Theme.AA_Menu_Drawer1" />
</resources>
```

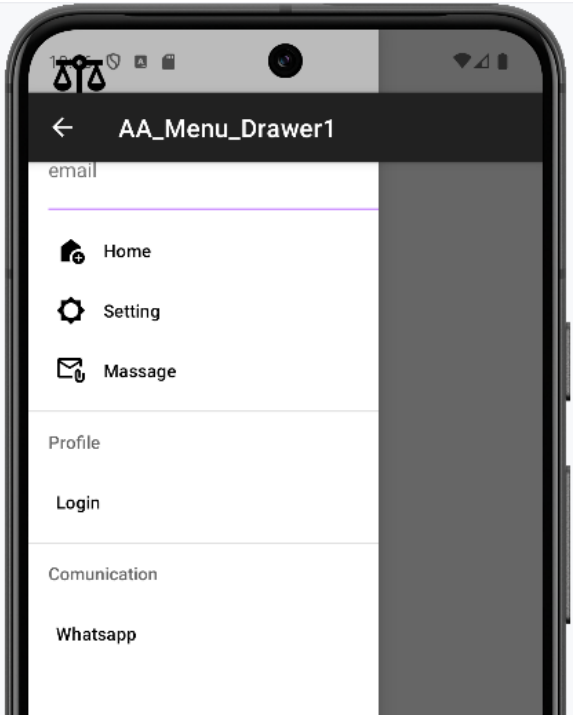


MainActivity.kt themes.xml

```xml
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.AA_Menu_Drawer1" parent="Theme.Material3.DayNight">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.AA_Menu_Drawer1" parent="Base.Theme.AA_Menu_Drawer1" />
</resources>
```

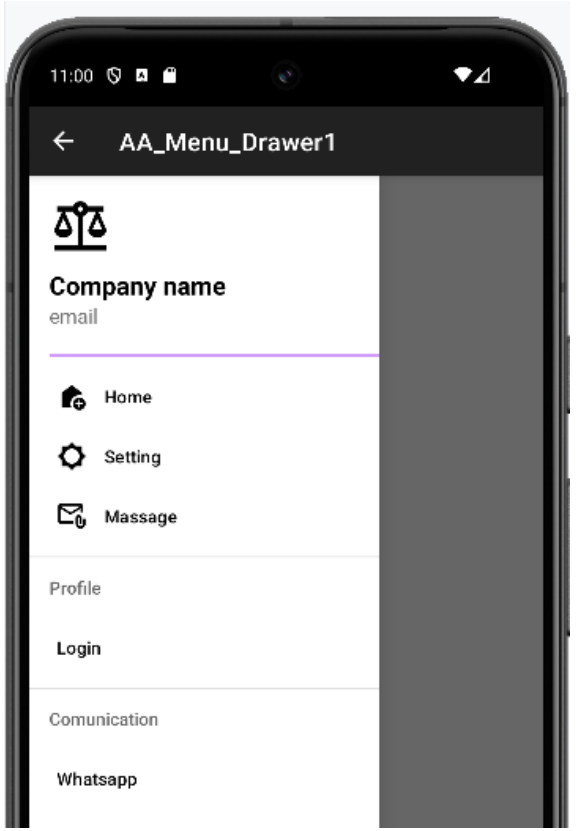


Test app




```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    enableEdgeToEdge()
    setContentView(R.layout.activity_main)
   drawerLayout = findViewById( id = R.id.drawerLayout)
   val navView : NavigationView = findViewById<Navigatio

    toggle = ActionBarDrawerToggle( activity = this,drawerLa
```

```
<com.google.android.material.navigation.NavigationView
    android:layout_width="wrap_content"
    android:layout_height="match_parent"
    android:layout_marginTop="?attr/actionBarSize"
    android:padding="20sp"
    app:headerLayout="@layout/nav_header"
    app:menu="@menu/nav_menu"
    android:id="@+id/nav_view"
    android:fitsSystemWindows="true"
    android:layout_gravity="start" >
</com.google.android.material.navigation.NavigationView>
```



Close Drawer Menu after clicked

```
navView.setNavigationItemSelectedListener {
    when(it.itemId){
        R.id.nav_home -> Toast.makeText( context = this,  text =
        R.id.nav_setting -> Toast.makeText( context = this,  te
        R.id.nav_login -> Toast.makeText( context = this,  text

    }
    drawerLayout.closeDrawers()
    true
}
```

Open LoginFragment inside MainActivity

1. Add new BlankFragment: LoginFragment

2. Change activity_main: add ContainerLayout (id=containerFragment)

```
    <androidx.fragment.app.FragmentContainerView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/containerFragment"
        android:textSize="30dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

3. Add these statement inside MainActivity

```
R.id.nav_login -> {
    supportFragmentManager.beginTransaction()
        .replace( containerViewId = R.id.containerFragment, fragment = LoginFragment())
        .commit()
}
```